

pyGuidos: A cross-platform Python interface to GuidosToolbox Workbench for pattern analysis of raster maps

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Software

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Summary

Forests and other natural habitats are increasingly threatened by fragmentation, degradation, and land-use change. Quantifying these processes requires spatially explicit indicators derived from land cover raster maps. The GuidosToolbox Workbench (GWB) is a free software developed at the European Commission Joint Research Centre (JRC) that provides command-line scripts to a suite of raster image analysis modules, including forest fragmentation assessment, landscape mosaic classification, pattern analysis, patch accounting, and network coherence metrics for automated processing on Linux x86_64 systems. The GWB modules rely on compiled binary executables and Interactive Data Language (IDL) wrapper routines. IDL is bundled as a standalone runtime, but compiling modified source code requires a proprietary license. The IDL software is also constrained to a limited set of operating systems and CPU architectures, hence compatibility with future platforms cannot be guaranteed.

pyGuidos is a Python package that provides a direct programmatic interface to the core GWB modules across different operating systems and CPU architectures. All spatial analysis functions are coded in Python using Numba ([Lam et al., 2015](#)) JIT-compiled kernels for high-performance parallel computation, eliminating all dependencies on external binaries and the IDL runtime. The current version (2.3.2) implements six landscape analysis functions and one temporal change detection function.

Statement of Need

From GUI and bash to native Python

Guidos is an acronym for Graphical User Interface for the Description of Image Objects and their Shapes. The [Guidos ecosystem](#) has evolved through three delivery mechanisms: first, the desktop application GuidosToolbox ([Vogt & Riitters, 2017](#)), or [GTB](#) (available for x86_64 of Linux, macOS, and Windows) with IDL bundled alongside various precompiled binaries; second, the command-line application GuidosToolbox Workbench ([Vogt et al., 2022](#)), or [GWB](#) (constrained to Linux x86_64); and third, [pyGuidos](#) v1, a Python wrapper around GWB bash calls inheriting the Linux-only constraint. These limitations prevented integration into Python-based scientific workflows or automated pipelines on non-Linux systems. More fundamentally, the IDL runtime creates a sustainability risk: GTB and GWB are already incompatible with arm64 versions of MS-Windows and Linux, and this incompatibility may worsen as platforms evolve.

pyGuidos v2.3 addresses these limitations by reimplementing all spatial engines as pure Python functions accelerated with Numba JIT compilation. The sliding-window operations run in

40 parallel across all CPU cores, achieving performance comparable to the original C binaries
41 while remaining fully portable. Input and output operate on individual GeoTIFF files using
42 rasterio, aligning with standard Python geospatial conventions.

43 Policy context: the EU Nature Restoration Regulation

44 The European Union Nature Restoration Regulation (NRR), which entered into force in August
45 2024, legally requires European Union Member States (MS) to report a forest connectivity
46 index based on the Foreground Area Density (FAD) method ([European Parliament and Council, 2024](#)). FAD measures the proportion of forest pixels within a moving window centered on
47 each forest pixel, providing a scale-dependent, per-pixel estimate of local forest connectivity
48 ([Riitters et al., 2002](#); [Vogt, 2025](#)). This legal mandate drives immediate demand for auditable,
49 cross-platform software capable of computing FAD at national and continental scales. Because
50 traditional GTB/GWB architectures rely on proprietary runtimes and platform-constrained
51 binaries, they are unsuitable for deployment in automated national reporting pipelines.
52

53 pyGuidos fills this gap directly: it provides the exact FAD algorithm adopted by the NRR,
54 implemented in pure Python code that can be deployed on any platform. This makes pyGuidos
55 a tool of direct practical relevance for environmental monitoring, with an audience extending
56 to national forest services and environmental agencies in all 27 MS and beyond.

57 Target audience

58 The package is intended for landscape ecologists building reproducible analysis chains, remote
59 sensing analysts applying GTB metrics to regional or global datasets, and forest monitoring
60 practitioners at institutions such as the FAO, national forest services, and European Commission
61 bodies. To ensure global practitioners can freely report issues and contribute code, public
62 community interactions are managed through an open [GitHub portal](#).

63 State of the Field

64 GTB is a widely used image analysis platform, with applications in forest monitoring, biodiversity
65 assessment, habitat connectivity, and land cover change detection ([Vogt et al., 2022](#); [Vogt & Riitters, 2017](#)). Its algorithms have been adopted in official reporting frameworks including the
66 FAO State of the World's Forests ([FAO and UNEP, 2020](#); [Vogt et al., 2019](#)), the EU MAES
67 ecosystem assessment ([Maes & others, 2020](#)), national and international forest monitoring
68 programs ([EUROSTAT, 2022](#); [FOREST EUROPE, 2026](#); [Vogt & Caudullo, 2025](#)).
69

70 pyGuidos is the first and currently only Python interface enabling GTB analyses on all major
71 platforms without external dependencies beyond standard Python packages.

72 The seven functions currently implemented and their methodological references are:

- 73 1. `frag()`: Forest fragmentation via Foreground Area Density (FAD) and Foreground Area
74 Clustering (FAC) methods ([Riitters et al., 2002](#); [Vogt, 2025](#))
- 75 2. `frag_change()`: Pixel-level fragmentation transition analysis across multiple time periods
- 76 3. `landmos()`: Classifies land cover heterogeneity within a tri-polar framework ([Riitters et al., 2009](#); [Vogt, 2022d](#); [Vogt et al., 2024](#))
- 77 4. `spa()`: Simplified Pattern Analysis, classifying foreground pixels into morphological
78 categories (Core, Edge, Perforation, Islet, Linear) ([Soille & Vogt, 2022, 2009](#); [Vogt, 2022b](#))
- 79 5. `acc()`: Patch accounting, labeling and classifying foreground patches into user-defined
80 area size classes ([Vogt, 2022a](#))
- 81 6. `rss()`: Restoration Status Summary, computing network coherence indicators including
82 Equivalent Connected Area (ECA) and coherence (COH) ([Saura et al., 2011](#); [Vogt, 2022c](#))
- 83
- 84
- 85

86 7. `extract_by_polygon()`: Zonal raster statistics extraction with automated CRS
87 reprojection

88 The SPA function implements a simplified version of the full Morphological Spatial Pattern
89 Analysis (MSPA) algorithm. While MSPA distinguishes up to 23 mutually exclusive
90 morphological feature classes, SPA provides up to 6 of the most used classes. A full native
91 Python implementation of MSPA is planned for future versions.

92 Software Design

93 Architecture

94 pyGuidos is structured as a pure Python package with six main analysis functions
95 (`fragmentation`, `fragmentation_change`, `land_mosaic`, `spa`, `accounting`, `rss`), alongside a
96 regional extraction utility (`extract_by_polygon`). It also includes modular submodules for
97 spatial processing (`engine`), computational utilities (`utils`), and data validation (`checks`).
98 To optimize performance, all computationally intensive operations leverage Numba `@jit`
99 decorated functions with `parallel=True` enabled for multi-core execution.

100 A key design decision was to operate on individual GeoTIFF files rather than directories. Each
101 analysis function validates inputs, delegates computation to `engine.py`, saves output GeoTIFFs
102 with embedded GTB metadata tags, and returns a nested dictionary containing output paths and
103 statistics. All output GeoTIFFs embed processing parameters in `TIFFTAG_IMAGEDESCRIPTION`
104 using a structured format, allowing standalone statistics functions to recover parameters from
105 output files without requiring the original function call context.

106 Native Spatial Engines

107 Reimplementing GTB modules as native Python functions yields significant architectural
108 benefits: absolute cross-platform portability without bundling compiled binaries, full algorithmic
109 transparency for scientific audit, and multi-core parallelism via Numba's thread-level prange
110 that matches optimized C performance. The SPA function combines `scipy` distance transforms,
111 `scikit-image` morphological reconstruction, and a Numba-parallelized assembly step to classify
112 foreground pixels into structural categories.

113 Regional Analysis Workflow

114 A scientifically important workflow involves computing a landscape metric over an entire study
115 area first — to preserve correct moving-window context — and then extracting per-region
116 statistics using `extract_by_polygon()`. This post-processing function clips output rasters to
117 polygons from any vector format supported by `pyogrio`, handling CRS transformations and
118 nodata masking automatically. This two-step approach ensures that metrics near administrative
119 boundaries are not biased by edge artifacts from adjacent areas outside of the administrative
120 boundary.

121 Validation and Testing

122 The package includes a comprehensive test suite covering all functions, using `pytest` with
123 mocking for I/O-intensive operations. The GitLab CI/CD pipeline runs the full suite on every
124 commit with code coverage reporting.

125 Repository and Community Access

126 The core development of pyGuidos is officially hosted by the European Commission on
127 code.europa.eu. Because access to the institutional forge requires an authorized EU Login,
128 a fully synchronized public mirror is maintained on [GitHub](https://github.com) providing an unrestricted issue

129 tracker and Pull Request workflow for the wider scientific community. A dedicated [online](#)
130 [manual](#) provides information on installation and usage of pyGuidos. The repository includes
131 reproducible example notebooks demonstrating the package's functions using the Copernicus
132 CORINE Land Cover 2000 and 2018 maps over Corsica ([European Environment Agency, 2020a](#),
133 [2020b](#)).

134 Research Impact

135 pyGuidos was developed as the computational backbone of the Global Forest Attribute Dataset
136 (GFAD), presenting global spatial layers and country-level statistics derived from the JRC
137 Global Forest Cover 2020 (GFC2020) map ([Caudullo & Vogt, 2026b](#)). The GFAD processing
138 chain leveraged pyGuidos to generate global, 100-meter resolution layers for fragmentation,
139 patch accounting, morphology, and restoration status. Processing this ~0.6 terapixel dataset
140 demonstrated operational scalability, completing the complex routine in 6.5 hours utilizing a
141 multi-core cluster infrastructure.

142 The GFAD dataset is integrated as a core component of the EU Observatory on Deforestation
143 and Forest Degradation (EUFO) and is aligned with the EU Nature Restoration Regulation and
144 international forest reporting frameworks. The underlying processing notebooks are publicly
145 archived to ensure full reproducibility ([Caudullo & Vogt, 2026a](#)).

146 JRC is also using pyGuidos as the basis for NRR-compliant training material for EU Member
147 States, with `frag_change()` directly supporting the temporal reporting requirement. This
148 adoption by national monitoring programs represents a direct pathway from research software
149 to policy-relevant operational use.

150 AI Usage Disclosure

151 Generative AI tools assisted in the development process: Anthropic's Claude aided in module
152 refactoring and paper drafting; Google's Gemini supported code generation, algorithm
153 optimization, and documentation structure; and Amazon's Kiro assisted in bug identification
154 and performance tuning for Numba kernels. All tools were utilized for writing unit tests and
155 code reviews.

156 All AI-generated content was reviewed, corrected, and validated by the authors. Core design
157 decisions, algorithm implementations, scientific methodology, and final form of all outputs are
158 the sole responsibility of the authors.

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163 methodology is based on the work of Soille and Vogt ([Soille & Vogt, 2022, 2009](#)).

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